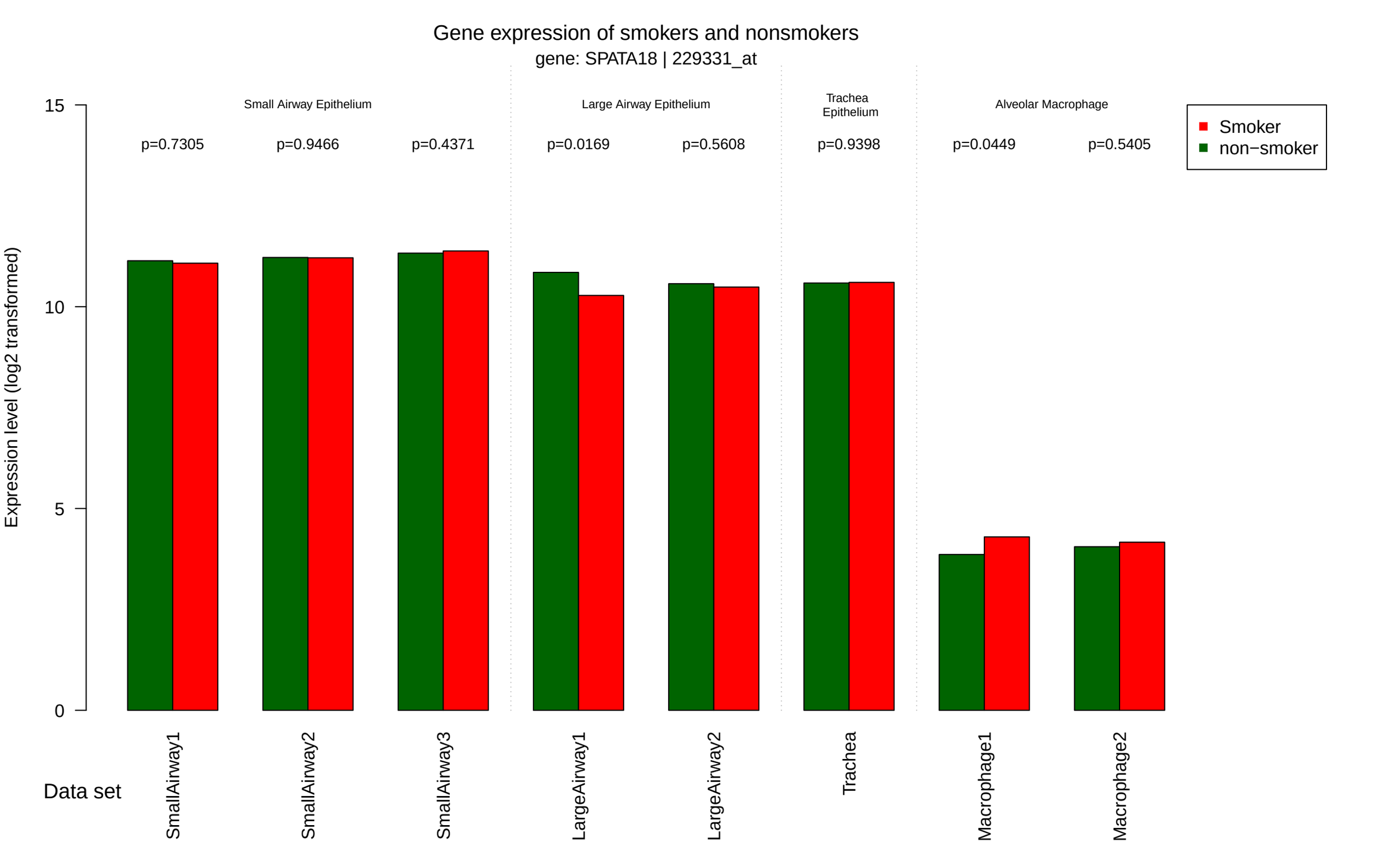
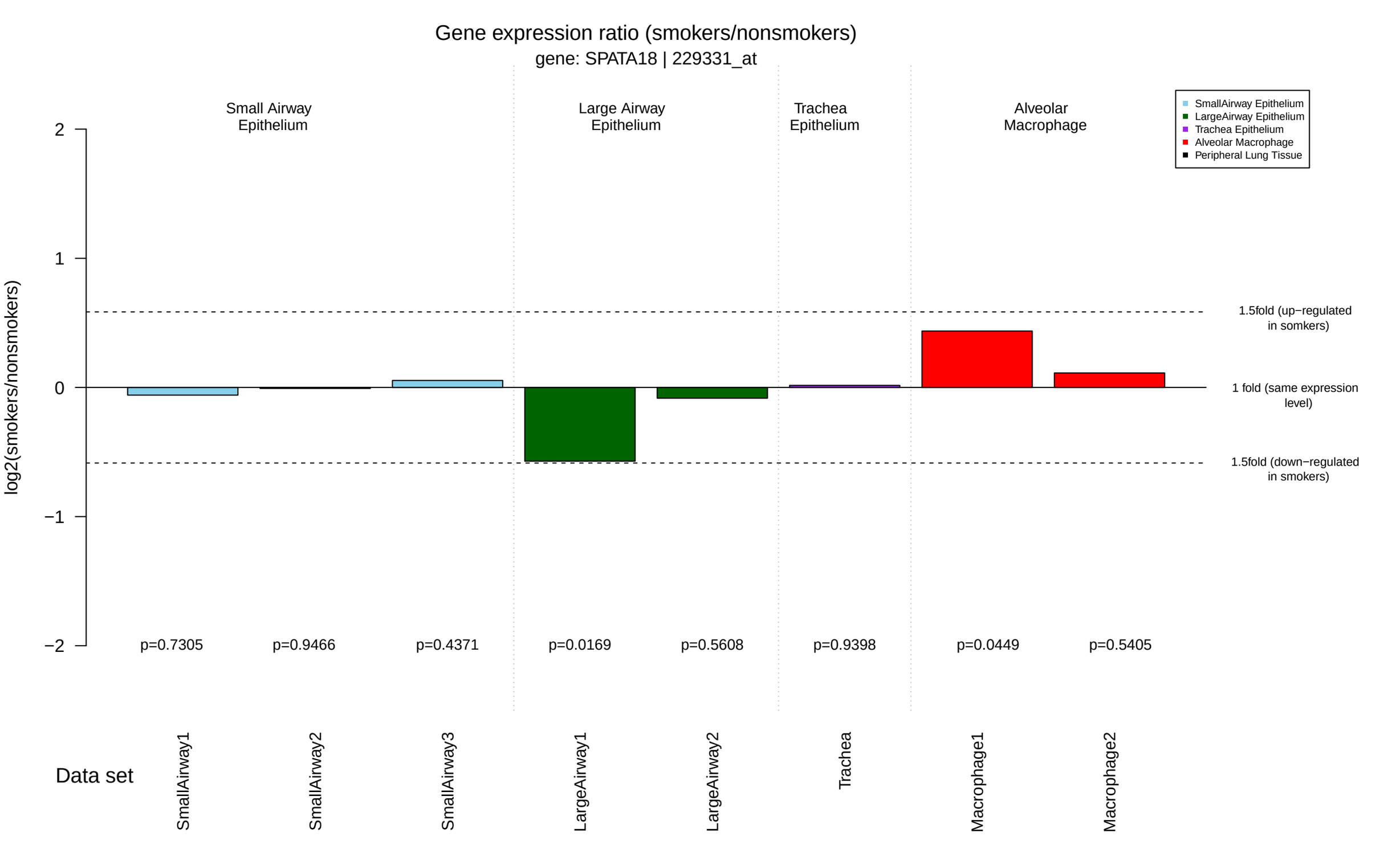
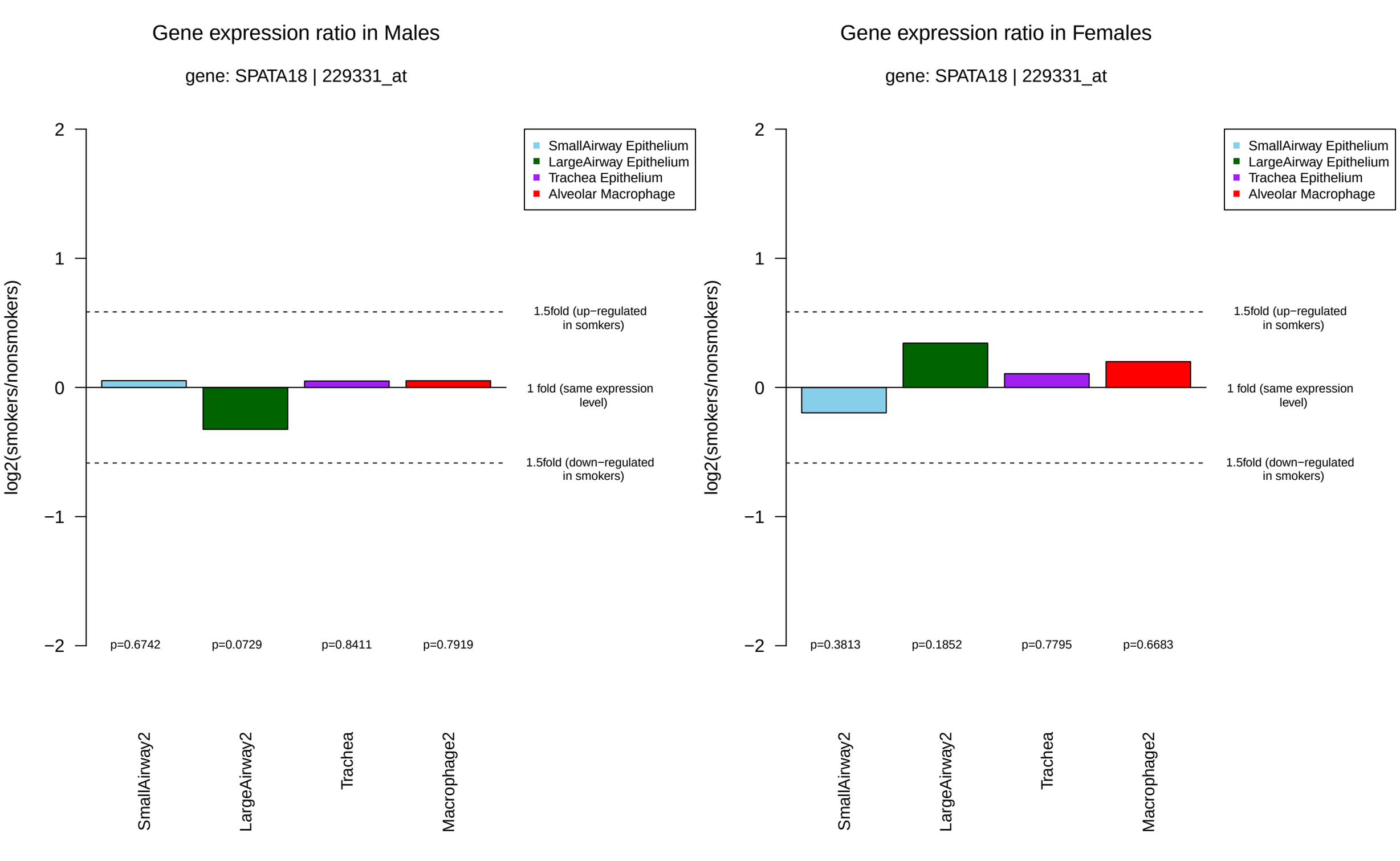
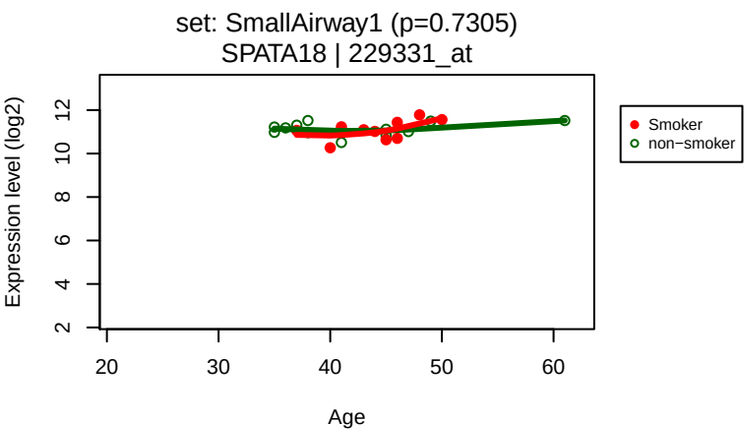
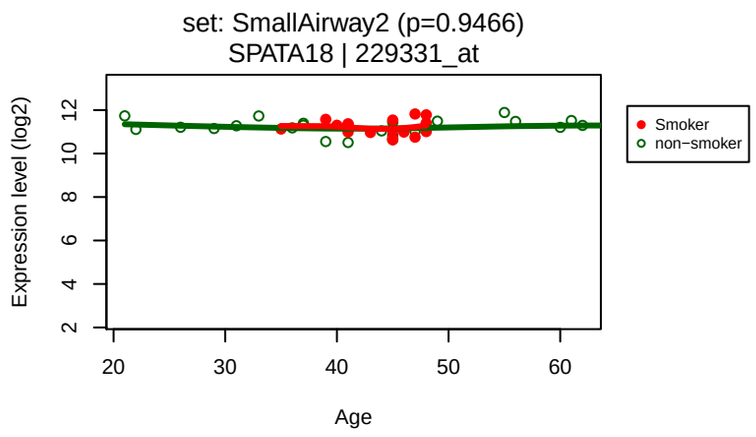
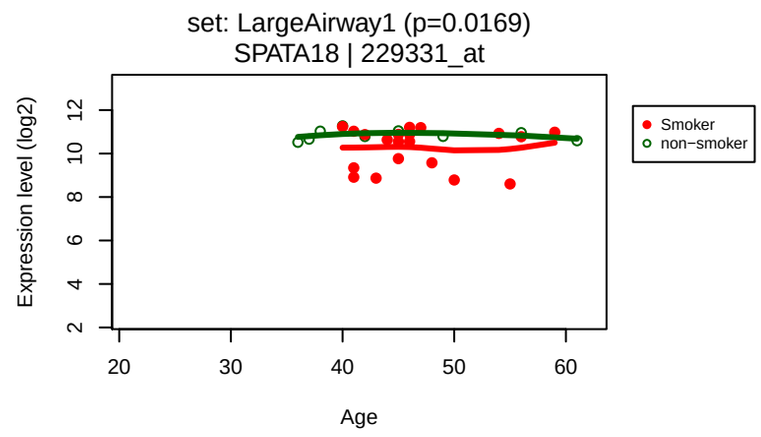

Gene expression of smokers and nonsmokers
gene: SPATA18 | 229331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7305
p=0.9466
p=0.4371
p=0.0169
p=0.5608
p=0.9398
p=0.0449
p=0.5405
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

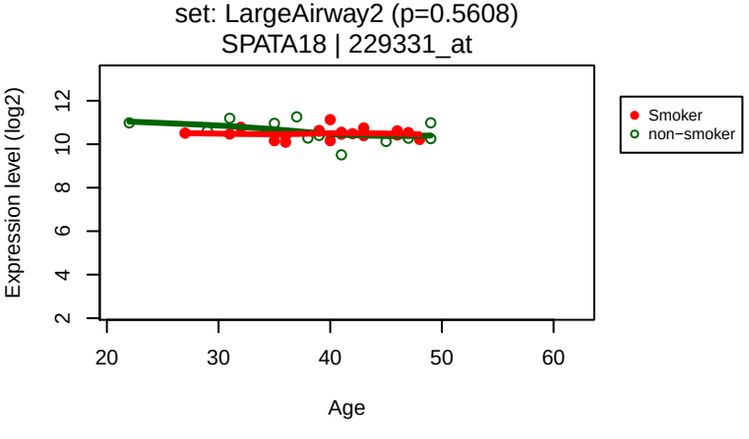

Gene expression ratio (smokers/nonsmokers)
gene: SPATA18 | 229331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7305
p=0.9466
p=0.4371
p=0.0169
p=0.5608
p=0.9398
p=0.0449
p=0.5405
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

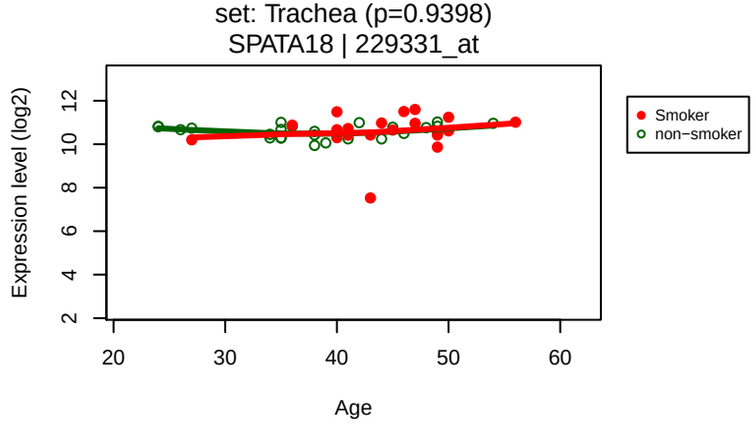

Gene expression ratio in Males
gene: SPATA18 | 229331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6742
p=0.0729
p=0.8411
p=0.7919
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA18 | 229331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3813
p=0.1852
p=0.7795
p=0.6683
SmallAirway2
LargeAirway2
Trachea
Macrophage2

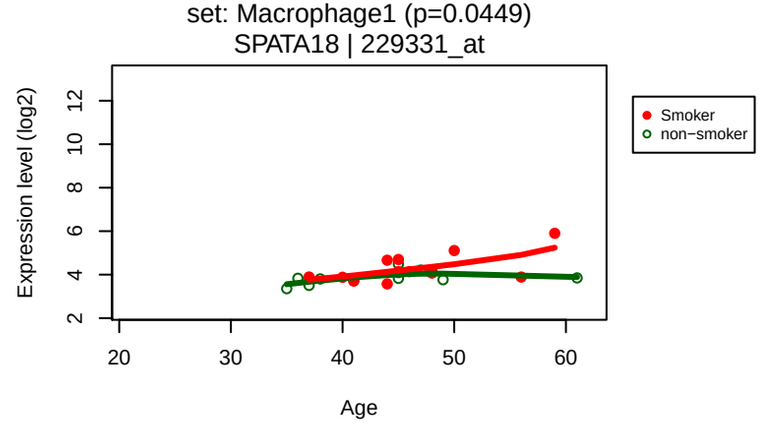

set: SmallAirway1 (p=0.7305)
SPATA18 | 229331_at
Expression level (log2)
Age
Smoker
non−smoker

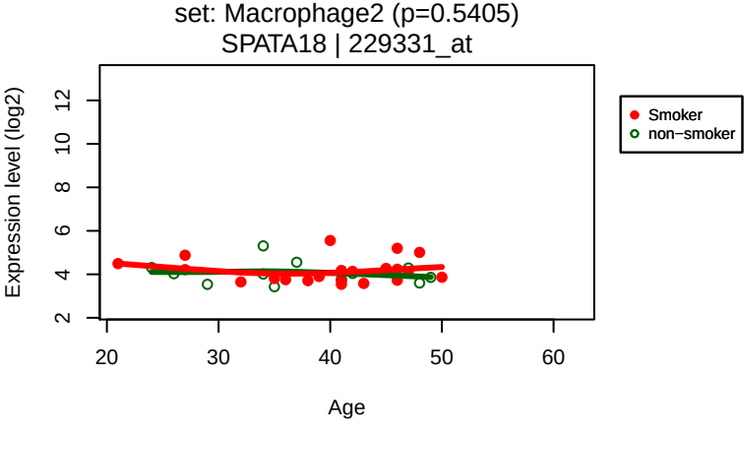

set: SmallAirway2 (p=0.9466)
SPATA18 | 229331_at
Expression level (log2)
Age
Smoker
non−smoker

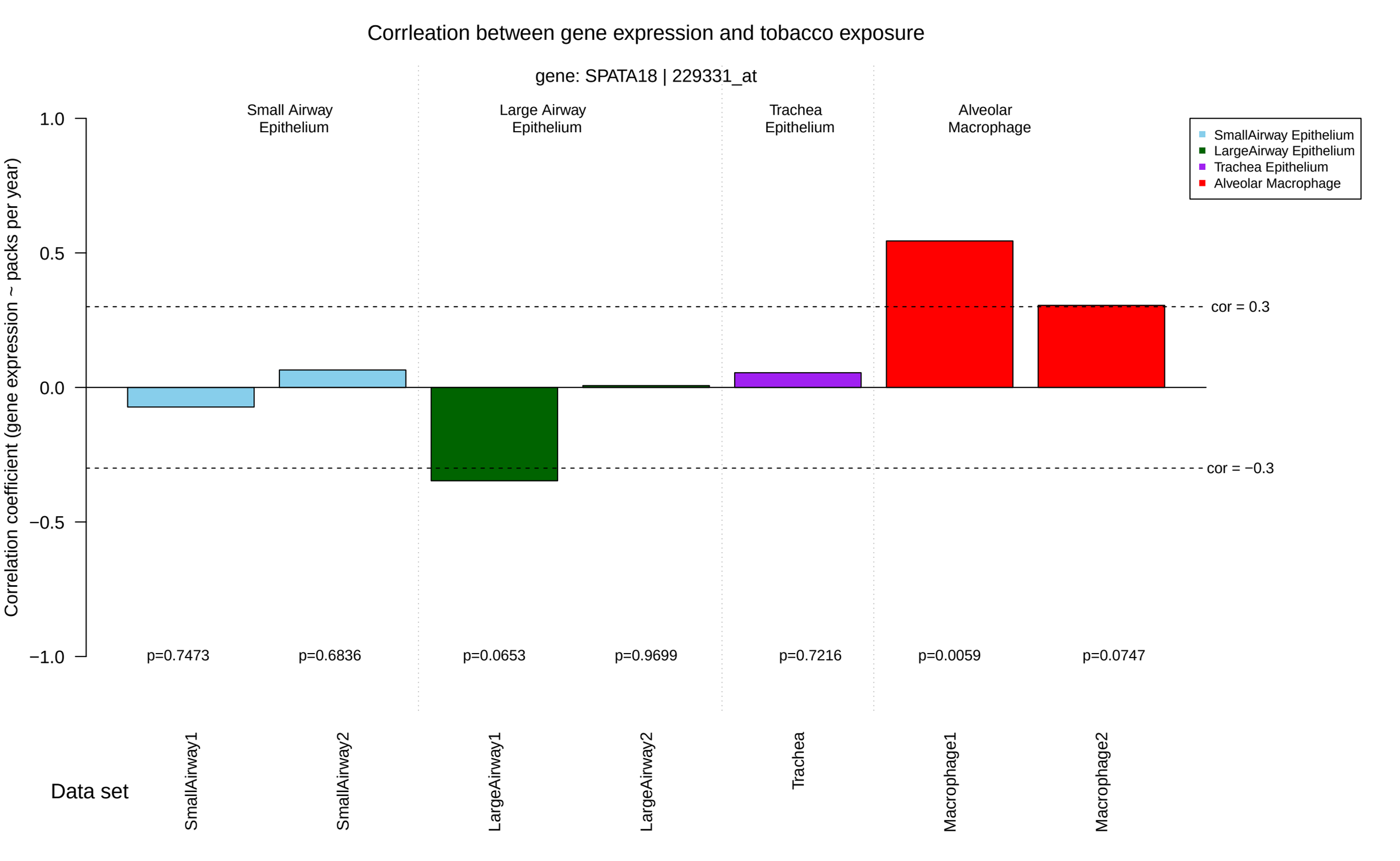

set: LargeAirway1 (p=0.0169)
SPATA18 | 229331_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5608)
SPATA18 | 229331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9398)
SPATA18 | 229331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0449)
SPATA18 | 229331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5405)
SPATA18 | 229331_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA18 | 229331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7473
p=0.6836
p=0.0653
p=0.9699
p=0.7216
p=0.0059
p=0.0747
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2